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Roll No.

TEE-701

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

ELECTRICAL DRIVES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) State and explain the function of various converters used in electric drives. (CO1, CO2)
- (b) Discuss the dynamic equation of electrical drive. (CO1, CO2)
- (c) State essential parts of electric drives. What are functions of power modulator ? (CO1, CO2)
2. (a) Draw and explain four quadrant operation of a drive system using suitable example. (CO3, CO4)
- (b) Why flywheel is employed for equalization of load derive the relation. (CO3, CO4)
- (c) Determine the fundamental torque equation used for motor load system. (CO3, CO4)

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3. (a) Draw and explain the circuit diagram for armature resistance control of separately excited dc motor. What are its limitations? (CO5, CO1)
- (b) With the help of suitable block diagram. Explain V/f control used in induction motor drives incorporating closed loop system. (CO5, CO1)
- (c) A motor drives two loads. One has rotational motion. It is coupled to the motor through a reaction gear with $a = 0.1$ and efficiency of 90%. The load has a moment of inertia of 10 kg-m^2 and a torque of 10 N-m . Other load has translational motion and consists of 1000 kg weight to be lifted up at an uniform speed of 1.5 m/s . Coupling between this load and the motor has an efficiency of 80%. Motor has an inertia of 0.2 kg-m^2 and runs at a constant speed of 1420 rpm . Determine equivalent inertia referred to the motor shaft and power developed by the motor. (CO5, CO1)
4. (a) Discuss regenerative and dynamic braking. What are drawback of dynamic braking? (CO3, CO6)
- (b) Discuss steady state stability limit. How is it used to decide stability of the system? (CO3, CO6)
- (c) A drive has following parameters :
 $J = 10 \text{ kg-m}^2$. $T = 100 - .01N$, N-m Passive load torque $T_r = 0.05 \text{ N, N-m}$, where N is the speed in rpm.
Initially the drives is operating in steady-state. Now it is to be reversed. For this motor characteristic is changed to $T = -100 - 0.1 \text{ N, N-m}$. Calculate the time of reversal. (CO3, CO6)
5. (a) Discuss various types of load characteristics in detail. (CO2, CO6)
- (b) How is plugging applied on motor? Draw the circuit diagram and explain? (CO2, CO6)
- (c) Explain closed loop control of CSI and VSI drives. (CO2, CO6)

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TEE-702

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

**NCER (NON-CONVENTIONAL RESOURCES AND ENERGY
SYSTEMS)**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each question carries 10 marks.

1. (a) What is the importance of MPPT in an SPV system ? Name the various strategies used for operation of an MPPT. (CO1, CO2)
- (b) Explain the working of distributed collector solar thermal electric power plant. (CO1, CO2)
- (c) Explain the flat plate solar collector. How is the collection of solar energy affected by tilting a flat plate solar collector ? (CO1, CO2)
2. (a) With the help of a schematic diagram, explain the working of solar water heating. (CO1, CO2)

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- (b) Explain the I-V characteristics of solar cell and define fill factor. What is significance of fill factor ? (CO1, CO2)
- (c) With the help of block diagram explain the operation of standalone and grid interactive solar generation plant. (CO1, CO2)
3. (a) Describe the classification of the fuel cells. (CO3, CO4)
- (b) Describe various fuels used in fuel cells along with the chemical reaction involved. (CO3, CO4)
- (c) What are the main hurdles in the way of common use of fuel cell and also the main application of fuel cells. (CO3, CO4)
4. (a) What are the factors responsible for the distribution of wind energy on the surface of earth ? (CO5)
- (b) With the help of a diagram, explain the terms : free and relative wind velocity, drag and lift forces, solidity, pitch angle and chord. (CO5)
- (c) Explain the variation of power output of a wind turbine with tip speed of the rotor. (CO5)
5. (a) Describe the basic principle of operation of an MHD generator. Derive expression for maximum power generation per unit volume of generator. (CO4, CO6)
- (b) What are the main advantages and disadvantages of biomass energy ? (CO4, CO6)
- (c) Explain with the diagram of open cycle MHD generation system. (CO4, CO6)

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TOE-703

**B. TECH. (EEE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

BIOMEDICAL ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is propagation of Action Potential ? (CO1, 2)
(b) Elaborate the basic design specifications of medical instrumentation systems. (CO1, 2)
(c) Define electrodes and bio-potential electrodes with the help of the diagram. (CO1, 2)
2. (a) Briefly explain the types of electrodes that are used to measure bioelectric events. (CO2)

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- (b) What are force transducers, pressure transducers and transducers using digital output ? (CO2)
- (c) Briefly explain the passive transducers and its types. (CO2)
- 3. (a) Write a short note on ECG and EMG. (CO3)
- (b) How could we measure the blood flow and cardiac output ? (CO3)
- (c) Explain cardiovascular measurements with the help of the diagram. (CO3)
- 4. (a) Explain about inhalators, respirators and ventilators in detail. (CO4)
- (b) Give a brief description of spirometer used in breathing measurements. (CO4)
- (c) Describe about physiology of respiratory systems. (CO4)
- 5. (a) Describe about principal of X-ray measurements. (CO5)
- (b) With the help of the diagram explain properties of ultrasound, also define amount of frequency shift. (CO5)
- (c) Discuss about CAT scan in detail. (CO5)

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TEE-704

**B. TECH. (EE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

FUNDAMENTALS OF IoT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Ericsson, Inc. of Plano, TX, is a leading provider of technology and telecom services for mobile and fixed networks. Their services include communication systems, IoT implementation, device integration, and more, and their IoT Accelerator enables customers to build cellular IoT solutions for any data-driven application. Design an IoT architecture in the perspective of Ericsson. (CO1)
- (b) Samsara's sensor data solutions aim to enhance operational safety and efficiency for an array of industries. The company's connected sensors allow businesses to more easily manage, track and monitor operations within vehicles and fleets. Classify the sensors used in Samsara's sensor data solutions. (CO1)

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- (c) Digi International Ltd. is a manufacturer and service company that focuses on embedded systems, RF modules, internet infrastructure, and other wireless technologies. They sell a suite of Digi embedded systems that allow modular conversion of existing hardware to securely connect to their IoT services. Sketch the WSN architecture in the perspective of Digi International Ltd's IoT platform. Also, describe the role of WSN in manufacturer company. (CO2)
2. (a) Amtech Electrocircuits, Inc. of Troy, MI, is a Thomas Registered manufacturer of electronic wire harnesses, PCBs, mechanical assemblies, and more. They provide engineering, testing, prototyping, and other related services on their products. Their capabilities can be used to create IoT connectivity through their high-quality electronics and customer-based manufacturing methods. Summarize the benefits of IoT in the perspective of Amtech Electrocircuits. (CO2)
- (b) SBT Alliance is an engineering; service company that deploys, integrates, and supports IoT lighting solutions for industrial applications. They work with energy efficiency system to reduce operating costs and leverage smart lighting devices to create smart spaces for their customers. Apply the Key IoT Technologies and design a framework for SBT Alliance. (CO3)
- (c) Hyperlink InfoSystem delivers the best Internet of Things apps to support smart initiatives from the various industries ranging from security to home automation to industrial IoT that can change business operations and change people's lives. Identify the quality of services parameters for wireless sensor network in the perspective of Hyperlink InfoSystem. (CO3)

3. (a) Electric car charging is fast-becoming a competitive industry. The number of electric car charging companies on the kerbside is rapidly increasing and fossil fuel giants like Shell and BP are buying up established networks to expand their own offering. Use the components of RFID to design electric car charging stations. (CO3)
- (b) Larsen & Toubro Infotech Limited is an Indian global information technology services and consulting company based in Mumbai, India. LTI helps Hi-Tech companies innovate faster and operate in real-time by leveraging IoT for vast business outcomes, including predictable yield and optimized cost. Design a IoT protocol stack in the perspective of Larsen and Toubro Infotech Limited. (CO4)
- (c) At Mphasis, they develop a system for IoT-enabled block chain-based automated claims administration in the insurance sector. With the use of block chain technology, the problem detection, repair, and claims administration processes for IoT devices may all be automated. Construct a security architecture of IoT with its Security requirements in the perspective of Mphasis. (CO4)
4. (a) Contec Americas Inc. is a Thomas Verified manufacturer and service company of embedded devices, software, controllers, and more, based out of Melbourne, FL. They sell industrial IoT, M2M, and factory automation systems to all seven continents and have integrated over 50,000 IoT systems in their 40 years of service. Classify the identity management models in the perspective of Contec Americas Inc.

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- (b) Softdel, based in Stamford, CT, is an IT service provider for IoT applications such as smart buildings, automation, and other IoT solutions. They will design, consult, develop, and integrate IoT functionality into your workspace, and they have successfully worked with Fortune 500 companies; over two decades to deliver quality solutions. Evaluate the clustering Principles in an Internet of Things Architecture in the perspective of Softdel. (CO5)
 - (c) Softua, Inc. is a manufacturing and service company of software, IT consulting, AI & machine learning, business intelligence/analytics, could enablement, and more. Their IoT services work alongside companies to build, store, connect, and manage devices over their IoT platform to convey savings and increased efficiencies to their customers. Compare the user-centric identity management, device centric identity management and hybrid-identity management in the perspective of Softua. (CO5)
5. (a) VVDN has been developing software for more than ten years. Their major goal is to assist international clients with their upcoming technological innovations in the fields of 5G, wireless, AI/ML on vision based solutions, IoT, cloud, and apps. Identify the security requirements of basic IoT applications. (CO5)
- (b) Smart waste management is characterized by the usage of technology in order to be more efficient when it comes to managing waste. Identify the use cases and misuse cases in the perspective of waste and recycling services of smart waste management (CO6)
 - (c) Telit offers the industry's best IoT products and solutions. This firm helps you to provide the best IoT expertise and development resources Telit offers products and solutions that include cellular communication modules in all technologies, GNSS, short-to-long range wireless modules, and JoT software platforms. Illustrate the layered attacker model in the perspective of Telit. (CO6)

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TEE-706

**B. TECH. (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

UTILIZATION OF ELECTRICAL ENERGY

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each question carries 10 marks.

1. (a) Define the following : (CO1)
(i) Luminous Flux
(ii) Luminance Intensity
(iii) Illumination
(b) Discuss the laws of illumination and its limitation in practice. (CO1)
(c) A drawing hall 40 m × 25 m × 6 m high is to be illuminated with metal-filament gas-filled lamps to an average illumination of 90 lm/m² on a working plane 1 metre above the floor. Estimate suitable number, size and mounting height of lamps. Sketch the spacing layout. Assume

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coefficient of utilization of 0.5, depreciation factor of 1/1.2 and spacing/height ratio of 1.2 : (CO1)

Watts	Luminous efficiency (lm/W)
200	16
300	18
500	20

2. (a) Discuss the various modes of heat dissipation. (CO2)
- (b) A 20-KW, 1-phase, 220-V resistance oven employs circular nickel chrome wire for its heating elements. If the wire temperature is not to exceed 1107°C and the temperature of the charge is to be 500°C , calculate the length and size of wire required. Assume a radiating efficiency of 0.6 and the specific resistance of nickel chrome as $100 \times 10^{-6} \Omega \text{ cm}$ and emissivity 0.9. (CO2)
- (c) Explain the principle of induction heating. What are the applications of induction heating ? (CO2)
3. (a) Differentiate between fusion and non-fusion welding. Further classify the welding process into various categories. (CO3)
- (b) Discuss the importance of Arc Welding machines. What are the advantages of coated electrodes ? (CO3)
- (c) Explain the process of carbon arc welding. (CO3)
4. (a) What are the advantages and disadvantages of electric traction ? (CO4)
- (b) Discuss the various systems of Railway Electrification. (CO4)

(3)

- (c) A train is required to run between two stations 1.5 km apart at a schedule speed of 36 km/h, the duration of stops being 25 seconds. The braking retardation is 3 km/h/s. Assuming a trapezoidal speed/time curve, calculate the acceleration if the ratio of maximum speed to average speed is to be 1.25. (CO4)
5. (a) Discuss the General Features of a Traction Motor. Explain the terms tractive effort, coefficient of adhesion, train resistance and specific energy consumption of train. (CO5 & CO6)
- (b) Explain the types of Braking used in Traction. (CO5 & CO6)
- (c) Discuss why a D. C. series motor is ideally suited for the traction services. (CO5 & CO6)